

VATSA JOSHI

AI / ML Engineer • LLM & Agentic Systems • Generative AI

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SUMMARY

AI/ML engineer specializing in LLM and agentic systems, with production experience building multi-agent pipelines, RAG/retrieval architectures, and fine-tuned models. Shipped a legal-AI NER pipeline that cut extraction cost 25× across an 8M-judgment corpus, and delivered multi-agent products end-to-end from data ingestion to deployment. IEEE-published and hackathon-winning, open to relocation across India and internationally.

TECHNICAL SKILLS

LLM & Agents: LangChain, LangGraph, CrewAI, FastMCP, RAG, CAG, LoRA/QLoRA, Unsloth, FAISS, Qdrant, Groq, prompt engineering

AI / ML: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace Transformers, DeepSpeed, fine-tuning, model training

Vision: OpenCV, YOLO, SSD, PaddleOCR, Diffusers

Languages: Python, JavaScript / TypeScript, Rust

Backend & Cloud: FastAPI, NestJS, Node.js, Docker, AWS, PostgreSQL, MongoDB, Elasticsearch, GraphDB, Redis

Frontend: Next.js, Astro, React Native

WORK EXPERIENCE

Founding AI Engineer • Maaya AI

Apr 2026 – Present

- Building Maaya's memory layer — an AI that recalls a user's meetings, emails, tasks, and commitments better than they can — owning the retrieval and memory infrastructure end-to-end.

Software Engineer Trainee (AI) • Promact, Vadodara

Jan 2026 – May 2026

- Contributed to **BharatLaw.ai**, an AI-powered legal-tech product for Indian law firms.
- Reduced legal NER extraction cost **25×** (₹1 → ~4 paisa per document) on an 8M-judgment corpus by replacing a 7-stage LLM pipeline with a 2-call Gemini-2.5-Flash-Lite architecture.
- Achieved **96% document-level zero-failure** across 457 test rows using DB/CSV-grounded structural extraction and windowed precedent extraction.
- Built AI-driven document workflows for translation, parsing, indexing, and retrieval.
- Developed multi-agent legal systems for petition drafting, judge profiling, litigation strategy, lawyer recommendations, and compatibility analysis.

Full Stack Developer (Internship) • Hiraeth Tech, Vadodara

Jun 2025 – Dec 2025

- Built scalable products with NestJS, Next.js, and PostgreSQL, including API integrations and version-controlled deployments.

Deep Learning Engineer (Internship) • Prodigious AI, Remote

Mar 2025 – May 2025

- Trained CNN image classifiers reaching **92% accuracy**; fine-tuned transformer models for NLP tasks.
- Built real-time object-detection pipelines with YOLO and SSD; contributed to LLM architecture for a finance firm.

AI/ML Engineer (Internship) • Sparks To Ideas, Remote

Feb 2025 – Apr 2025

- Trained an XGBoost customer-churn model reaching **95% accuracy**.
- Built an AI content-intelligence pipeline with LLM parsing and FAISS HNSW multilingual deduplication.

SELECTED PROJECTS

- PersonAI** (Groq, LangChain, FastMCP, Next.js) — Agent automating Gmail, Calendar, and Tasks via Workspace APIs with OAuth 2.1.
- FloWork** (LangGraph, Groq, Next.js, Flask) — n8n-inspired agentic workflow-automation platform with a visual node builder and AI conditional routing.
- LegalOCR** (PaddleOCR, FastAPI, Elasticsearch) — Legal document parser with 22+ language OCR, semantic chunking, and Elasticsearch indexing.
- Hindi2Guj-Qwen3.5** (Qwen, DeepSpeed, Accelerate) — Fine-tuned Qwen3.5-0.8B for Hindi-to-Gujarati translation using ZeRO distributed training.
- Rustformer** (Rust, Transformers) — Transformer architecture implemented from scratch in Rust for inference research.
- Deepfake Video Detection** (Deep Learning, Vision) — EfficientNetB7 + Bi-LSTM detector; 2nd place at Tinkerthon 3.0 with a 1-year SSIP Gujarat research grant (IEEE-published).

EDUCATION

B.Tech in Computer Science • Navrachana University, Vadodara

Aug 2022 – May 2026

PUBLICATIONS & AWARDS

- EfficientNetB7 & Bi-LSTM for Deepfake Detection** — IEEE Xplore (DOI: 10.1109/CCIC68129.2026.11486188).
- Report & Analysis of the Rabin-Karp Algorithm; Blockchain Technology — ResearchGate (DOIs: 10.13140/RG.2.2.24592.01283, 10.13140/RG.2.2.32465.75365).
- 2nd Place** — **Tinkerthon 3.0** + 1-year SSIP Gujarat grant; **Top 10** — Odoo x NMIT Hackathon.